

GymOS

Product Requirements Document

V1.5 — Beta-Ready Release

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Scope	V1.5 — Beta-Ready
Extends	V1.0 PRD (v0.4)

Convention: stable requirement IDs (FR-NNN, NFR-NNN) are never renumbered. V1.0 ended at FR-086 / NFR-010; V1.5 begins at FR-087 / NFR-011. Amendments reference the original FR and state the change rather than editing V1.0 text.

1. Overview

V1.5 turns the V1.0 pilot into a beta the founding team can put in front of real gyms and their real members with confidence. V1.0 proved the retention spine — payments, attendance, the front-desk alert, multi-tenant isolation. V1.5 adds the pieces that make a gym's day-to-day genuinely runnable without founder hand-holding, and the member-facing depth that gives people a reason to open the app between visits.

Three things define this release:

- **Complete client profiles with body & progress tracking.** Members log weight, measurements, and progress photos and see their trend over time — a reason to open the app on a rest day. Private by default; visible to an assigned coach only when one exists.
- **Self-serve staff management.** V1.5 lets a gym Owner add and manage their own Managers, Receptionists, and Coaches from the dashboard — the precondition for a gym operating on its own during a beta.
- **Tara Money as the automated payment option.** V1.5 integrates Tara Money (Cameroon mobile-money, developer API) as the automated way to pay, offered alongside cash and manual methods — not a replacement for them. It uses the same PaymentProvider interface built in V1.0; Notch Pay remains a documented fallback. Integration is gated by a sandbox spike with the same exit criteria that gated Notch Pay.

Alongside these, V1.5 delivers the roadmap features already sequenced for it — workout plans, class scheduling, quiet-gym alerts, class reminders — and completes the WhatsApp (Evolution API) member-invite and OTP-fallback work carried over from V1.0.

2. What Changed Since V1.0

Area	V1.0	V1.5
Automated mobile-money provider	Notch Pay	Tara Money (alongside cash; Notch Pay = fallback)
Member payment routing	Implicit	Into each gym's own Tara Money account
Gym SaaS billing	Tier assigned, not collected	Automated recurring (gym owner -> GymOS)
Tenant suspension	Manual (Super Admin)	Automated on SaaS non-payment, after grace
Client profile	Name, phone, plan, status	+ body metrics, measurements, photos, trends
Staff accounts	Super Admin creates Owner only	Owner self-serve: Manager, Receptionist, Coach
Coach data access	Assigned members + notes	+ member progress data (when assigned)
App between-visit value	Status, history	+ progress tracking, quiet-gym alerts
Classes	Deferred	Scheduling + booking + reminders
Workout plans	Deferred	Coach-authored plans
WhatsApp invite / OTP fallback	Started, unfinished	Completed
Product analytics	Sentry only	+ PostHog
Test automation	Manual + RLS/payment CI	+ E2E automation baseline

3. Goals & Success Metrics

3.1 Goals

#	Goal
G-7	Ship a beta a gym can operate end-to-end without founder involvement — including provisioning its own staff
G-8	Give members a reason to open the app between gym visits, measured by non-visit-day app opens
G-9	Complete the Notch Pay -> Tara Money cutover with zero lost or double-charged payments
G-10	Keep member body/progress data private by construction — no cross-member and no unauthorized coach access

#	Goal
G-11	Prove classes and workout plans work in a real gym's weekly rhythm without adding operational load

3.2 Success Metrics

Metric	Target	Counter-Metric
Beta gym self-sufficiency	>= 80% of beta gyms complete a full week with zero founder support tickets	Founder-intervention rate per gym/week
Non-visit-day app opens	>= 20% of active members open the app on a non-visit day within 60 days	Progress-logging abandonment after first entry
Payment cutover integrity	Zero lost or duplicated payments across migration	Reconciliation discrepancies during cutover
Progress-data isolation	Zero unauthorized reads of member body data	Coaches wrongly blocked from assigned members
Staff-provisioning safety	Zero privilege-escalation incidents	Legitimate staff-creation actions blocked
Class booking reliability	>= 95% of bookings honoured (no overbooking)	No-show rate (informational only)

4. Users & Roles

The V1.0 role hierarchy is unchanged. V1.5 makes the already-defined staff roles operationally real by giving the Owner the means to create and manage them, and extends the Coach's data access to include progress tracking. Role enforcement remains at the PostgreSQL RLS layer (FR-003).

4.1 Role capability deltas in V1.5

Role	New in V1.5
Owner	Create, edit, deactivate, reassign Managers, Receptionists, Coaches; assign members to coaches
Manager	Manage classes and schedules; cannot manage staff accounts or settings
Receptionist	Manage class bookings at the desk; check members into classes
Coach	Author workout plans; view assigned members' progress data; manage class sessions they lead
Member	Log body metrics and progress photos; book classes; opt into quiet-gym alerts; follow a workout plan

5. User Journeys

UJ-6: The Owner staffs their gym

Grace's gym just joined the beta. She opens Settings -> Staff, clicks Add staff, adds her receptionist Aicha (name, phone, role -> Create). Aicha gets an SMS with a temp password and dashboard link. Grace adds her coach Emmanuel, who on login sees only the Coach Portal. Grace never contacts support — the gym is staffed in four minutes.

UJ-7: Amara tracks her progress

On a rest-day evening Amara opens the app, taps Progress, logs her weight and waist measurement, and adds a photo stored privately. The app shows her weight down 2.4 kg since joining. She didn't train today but she opened the app — exactly the behaviour V1.5 is built to create.

UJ-8: Emmanuel coaches with real data

Emmanuel opens an assigned client's profile and sees their goal, experience level, and now their progress trend. He writes a note about the plateau and adjusts the workout plan, swapping two exercises. The member sees the update on next app open. A member he isn't assigned to is invisible — including all their progress data.

UJ-9: Nadia schedules a week of classes

Nadia (Manager) creates a recurring HIIT Tue/Thu 6 PM class, capacity 15, coach Emmanuel. Members book from the app; when it fills, booking closes. An hour before each session booked members get a reminder push. The receptionist sees who's booked and checks them in.

UJ-10: The payment provider cutover

The team completes the Tara Money sandbox spike and records it in docs/decisions.md. On cutover day, new payments route to Tara Money; Notch Pay payments still processing reconcile under the old provider. No member is charged twice; reconciliation shows zero discrepancies. The adapter swap touched one config surface; payment logic did not change.

6. Functional Requirements

New requirements begin at FR-087. V1.0 requirements (FR-001–FR-086) remain in force unless a V1.5 requirement explicitly amends them.

6.1 Staff Management (Owner Self-Serve)

FR-087 — A gym Owner can create staff accounts for their own gym: Manager, Receptionist, and Coach. Captures full name, phone (E.164), and role. The Owner cannot create another Owner or a Super Admin. A Manager cannot create staff at all. Staff creation is audit-logged.

FR-088 — A newly created staff member is provisioned like V1.0 owner activation: an SMS with a temporary password and dashboard link. First login requires setting a new password. Until first login the account is `pending_activation`.

FR-089 — An Owner can edit a staff member's name and role, and deactivate (soft-delete) a staff account. Deactivation revokes access on the next request and is audit-logged with a mandatory reason. A deactivated Coach's notes and plan authorship are retained, visible to Owner and Manager.

FR-090 — Role changes take effect on the staff member's next authenticated request. Because role/gym claims are in the JWT (FR-003), the dashboard forces a token refresh or re-login after a role change so stale elevated claims cannot persist.

FR-091 — One phone number maps to one platform user (FR-001). A person may hold different roles at different gyms — each a separate binding. A person cannot hold two roles at the same gym; a new role replaces the prior binding (audit-logged).

FR-092 — A Coach account is a staff role for portal access. Assigning members to that coach remains a separate action on the Members page (FR-055). A Coach with no assignments sees an empty list with guidance to contact their Manager/Owner.

6.2 Complete Client Profiles — Body & Progress Tracking

FR-093 — The member profile is extended with an optional body profile: height, starting weight, and optional baseline measurements. Entered during an optional 'Complete your profile' step or any time from Progress. No body data is mandatory.

FR-094 — A member can log progress entries over time — any subset of: weight, body measurements (waist, chest, hips, arms, thighs), a progress photo, and a note. Each entry carries a timestamp and a `client_id` for offline-safe dedupe. A member may soft-delete their own entry.

FR-095 — Body and progress data is private by default. RLS visibility rules:

- A member can always read and write their own body/progress data.
- An assigned Coach can read the progress data of members currently assigned to them — and only those members.
- A Coach's access ends the instant their assignment ends (FR-055 ended_at).
- No other role — Receptionist, Manager, Owner, or another member — can read a member's measurements or photos.
- Progress photos: readable only by the owning member and, if the member opts in per-photo, their assigned coach.

FR-096 — The member Progress screen shows current weight and change since start, a weight trend chart, logged measurements with trends, a photo timeline (member-only unless shared), and a log-entry action. Charts view offline; logging offline queues the entry (FR-097).

FR-097 — Amendment to FR-061. V1.0 limited offline support to check-in. V1.5 extends offline queueing to progress entry logging, stored locally (SQLite) and synced on reconnect via `client_id` idempotency. No other flows are offline in V1.5.

FR-098 — In the Coach Portal, an assigned member's profile gains a Progress tab showing weight/measurement trends and shared photos. The coach can add a note but cannot edit or delete a member's progress entries. Unassigned members are invisible and unreadable.

6.3 Payments — Tara Money (Automated Mobile-Money Option)

FR-099 — Amendment to FR-034. V1.5 integrates Tara Money (a Cameroon mobile-money service with a developer API — create a collect, then detect payment via callback/status) as the automated mobile-money option, replacing Notch Pay in that role. It is offered alongside cash and the manual methods (FR-033), not as a replacement — the automated way to pay, beside cash at the desk and in the app. Uses the existing PaymentProvider interface; business logic unchanged. Notch Pay remains a documented fallback. Provider selection is configuration, not code.

FR-100 — The Tara Money integration is gated by a sandbox spike with the same exit criteria that gated Notch Pay (OQ-2): sandbox auth, payment initiation returns a reference, webhook received and processed, idempotency test passes. Outcome recorded in docs/decisions.md. If the spike fails, cutover does not proceed.

FR-101 — Webhook signature verification (NFR-002) is provider-specific. The handler verifies using the active provider's scheme. Tara Money verification is implemented and tested against sandbox before cutover. Invalid payloads are rejected with HTTP 401.

FR-102 — Cutover procedure:

- New payment initiations route to Tara Money.
- Notch Pay payments already in processing reconcile to a terminal state under Notch Pay; the reconciliation job polls both providers during the window.
- No payment is re-initiated across providers (prevents double-charge).
- The migration window and its reconciliation result are recorded in the audit log.
- Cutover is reversible by configuration for the duration of the beta.

FR-103 — Both MTN Mobile Money and Orange Money are supported via Tara Money (matching V1.0 coverage, FR-033). Cash, bank transfer, and manual mobile money remain first-class manual methods, unchanged.

6.3a Payment Gateway — Two Distinct Payment Flows

Two structurally different payment relationships. Both use the Tara Money integration (FR-099) but route money in opposite directions with separate lifecycles. New requirements FR-124–FR-138.

Overview: two flows, one adapter

	Flow A — Member pays Gym	Flow B — Gym Owner pays GymOS
Payer	Member	Gym Owner
Recipient account	The gym's own Tara Money account	The GymOS platform account
Purpose	Gym membership subscription	Platform SaaS tier (FR-073)
Amount set by	Gym Owner (plan prices)	Super Admin (tier prices)
Provider credentials	Per-gym (each gym connects its own)	Single, platform-level
On non-payment	Member loses gym access (FR-030)	Whole gym suspended after grace (FR-131)
Exists before V1.5	Yes (V1.0)	No — new in V1.5

The same PaymentProvider interface serves both; the difference is whose credentials the payment routes through. GymOS never holds member money (FR-124) and earns only the SaaS tier fee (FR-125).

Flow A — Member → Gym (membership payments)

FR-124 — When a member pays their gym by Tara Money, the payment settles directly into that gym's own Tara Money account. GymOS orchestrates (create collect, detect confirmation, reconcile, receipt) but never receives or holds member funds — a technical orchestrator over each gym's own account, not an aggregator. Paying by Tara Money is one option beside cash and the other manual methods (FR-033).

FR-125 — GymOS takes no commission on member->gym payments in V1.5. Platform revenue comes solely from the SaaS tier fee (Flow B). Provider transaction fees are borne by the gym (FR-039); GymOS neither absorbs nor marks them up.

FR-126 — Each gym must connect its own Tara Money account to collect automated mobile-money payments. Settings provides a 'Connect payment account' flow where the Owner authorizes their gym's merchant credentials. Credentials are stored encrypted, readable only by the payment service, never returned to any client, tenant-isolated.

FR-127 — Cash and Tara Money are co-equal payment options, not a primary and a fallback. A gym always takes cash and the manual methods (FR-033); connecting Tara Money adds the automated mobile-money option beside them. A gym without Tara connected loses no ability to operate — it collects by cash and manual entry as before, and the app simply does not surface the automated 'pay by Tara Money' action until the Owner connects. Connecting is an enhancement, never a prerequisite.

FR-128 — When a member initiates a mobile-money payment, the service resolves the gym's connected credentials (FR-126) and routes through them. A member never sees gym credentials. If credentials are missing/invalid/revoked, initiation fails gracefully directing the member to the desk, and the Owner is notified their connection needs attention.

FR-129 — Both member subscription purchase and renewal use Flow A. Renewal via the front-desk alert panel (FR-050) and self-service renewal from the app both route through the gym's connected account. The subscription lifecycle (FR-027–FR-032) is unchanged; only money routing is now explicitly the gym's own account.

Flow B — Gym Owner → GymOS (SaaS subscription billing)

FR-130 — GymOS bills each gym for its platform SaaS subscription per the gym's tier (FR-073) and interval (monthly/annual). New in V1.5: V1.0 assigned tiers and enforced caps (FR-086) but did not collect the fee. Billing is automated recurring collection via Tara Money into the platform's account.

FR-131 — A gym's platform subscription has its own lifecycle: active -> past_due -> grace_period -> suspended.

- active — SaaS fee paid; gym fully operational.
- past_due — a billing attempt failed; the gym stays operational and automated retries begin (FR-133).
- grace_period — retries exhausted; a Super-Admin-configurable window begins (default 7 days). Gym operational; Owner sees a 'renew to avoid suspension' banner and is notified.
- suspended — grace elapsed; the whole gym tenant is suspended (staff + members cannot log in), data retained. Payment restores full access. Formalizes the Super Admin suspend capability (FR-071).

FR-132 — Suspending a gym for non-payment suspends the entire tenant — every staff and member account loses access until the subscription is current. Reversible: a successful SaaS payment returns the gym to active and restores access immediately. Member states and all data are preserved through suspension.

FR-133 — Automated SaaS billing runs on each gym's billing anchor. On failure, the system retries on a defined schedule (default: 1, 3, 5 days after due) before moving to grace_period. Every attempt is recorded and the Owner notified (FR-135).

FR-134 — The Super Admin dashboard gains a Billing view: each gym's tier, interval, SaaS status, next billing date, last payment, failed attempts. Super Admin can mark a payment received (out-of-band), apply a credit/free period (beta gyms, FR-136), trigger a retry, or suspend/reactivate. All actions audit-logged (FR-080).

FR-135 — Gym Owners receive platform-subscription notifications distinct from member notifications: upcoming SaaS renewal, payment succeeded/failed (with retry date), entering grace, impending suspension. Owner-facing and mandatory (non-opt-out), as they concern continued platform access.

FR-136 — Beta accommodation. Super Admin can place a gym on a free or discounted plan (zero-price tier or credited period) so beta gyms aren't charged during validation. The billing machinery (FR-130–FR-135) is built and exercised, but whether a given beta gym is charged is Super-Admin policy per gym.

Shared requirements (both flows)

FR-137 — Both flows reuse the V1.0 integrity machinery: idempotent webhooks (FR-035), reconciliation (FR-036), append-only audit (FR-079), integer-XAF storage (FR-026). Flow B reconciles against the platform account, Flow A against each gym's account. Discrepancies in either are flagged.

FR-138 — The PaymentProvider abstraction carries a routing context identifying which account a payment belongs to (a specific gym for Flow A, the platform for Flow B). This context selects the correct credentials at initiation, verification, and reconciliation. Adding a provider or flow changes only the adapter and context, not the calling code.

6.4 Classes & Scheduling

FR-104 — A Manager or Owner can create classes: name, description, assigned coach, capacity, and a schedule (one-off or recurring). Classes are tenant-isolated like all other data.

FR-105 — Members with an active subscription on a class-eligible plan can book a class session from the app. Booking is capacity-limited; when full, booking closes ('This class is full'). Enforced server-side to prevent overbooking under concurrency.

FR-106 — A member can cancel a booking up to a gym-configurable cutoff (default 2 hours), freeing the spot. Booking and cancellation are not payments.

FR-107 — A Receptionist can view a session's booked members and mark attendance. Class attendance is distinct from floor check-in but uses the same member-status rules — an expired member cannot be checked in and triggers the front-desk alert (FR-049).

FR-108 — The member app shows upcoming booked classes on Home and a Classes screen of available/booked sessions. Workout plans and classes are separate features.

6.5 Workout Plans

FR-109 — A Coach can author a workout plan for an assigned member: a named plan with an ordered list of exercises (sets, reps, optional notes), created and edited from the Coach Portal.

FR-110 — A plan is assigned to exactly one member. The member sees their plan and can mark exercises/sessions complete; completion logging is offline-safe (`client_id`). Completion data is visible to the authoring coach.

FR-111 — If a coach assignment ends (FR-055), the previous coach's plan stays visible to the member and Owner/Manager but is not editable by a new coach until they take ownership — mirroring the V1.0 session-note handoff.

FR-112 — A shared exercise library (platform defaults; gym/coach can add custom entries, gym-scoped) backs plan authoring.

6.6 Quiet-Gym Alerts

FR-113 — A member can opt in to quiet-gym alerts (default off). When occupancy drops into the Low band (FR-047) during opening hours, opted-in members receive N-06 — the V1.5 activation of the reserved N-06 notification (FR-075).

FR-114 — Quiet-gym alerts are rate-limited: max 2 per day per member, min 3-hour gap, only during configured opening hours.

FR-115 — Quiet-gym alerts use the existing occupancy calculation (FR-046) — no new presence detection — preserving the honest-estimate guarantee.

6.7 Class Reminders

FR-116 — A booked member receives N-07 60 minutes before the session. Class reminders are opt-out (non-critical), per FR-076.

6.8 WhatsApp Invite & OTP Fallback (V1.0 Carryover Completion)

FR-117 — The Evolution API WhatsApp integration is completed, covering the two unfinished V1.0 stories. Member invitations (FR-082) can be sent via WhatsApp in addition to SMS, at the gym admin's choice.

FR-118 — OTP delivery uses a fallback chain: primary SMS with WhatsApp/Evolution API fallback when SMS fails. Transparent to the member; channel and outcome are logged for observability.

FR-119 — The Evolution API instance configuration (in-progress V1.0 Epic-1 story) is finalized: platform-level, managed by Super Admin, documented in docs/decisions.md.

6.9 Dashboard & App Additions

FR-120 — The Settings page gains a Staff section (Owner only) listing staff with role and status, plus Add/Edit/Deactivate (FR-087–FR-089). All other Settings capabilities unchanged.

FR-121 — A new Classes page (Manager for create/edit; Receptionist for bookings and class attendance) lists classes, sessions, booking counts vs capacity, and the assigned coach.

FR-122 — The Coach Portal gains Workout Plans and, per assigned member, a Progress tab. No other dashboard section becomes visible to the Coach role.

FR-123 — The member app gains a Progress tab and a Classes tab alongside Home/Check-In/Profile. Notification preferences gain N-06 and N-07 toggles.

7. Non-Functional Requirements

New NFRs begin at NFR-011. V1.0 NFRs remain in force.

NFR-011 — Progress photos are stored in Supabase Storage under access rules mirroring FR-095: retrievable only by the owning member and, if shared, their assigned coach. Object paths are non-guessable; no photo is served from a public bucket.

NFR-012 — The Tara Money cutover must produce zero double-charges and zero lost payments, verified by the reconciliation job reporting zero discrepancies before Notch Pay is stood down as primary.

NFR-013 — Staff provisioning must make privilege escalation impossible: no role can create a role equal to or above its own, enforced at the RLS/auth-hook layer. CI asserts an Owner cannot mint a Super Admin and a Manager cannot mint staff.

NFR-014 — PostHog product analytics is integrated on app and dashboard (activating deferred NFR-008), focused on the V1.5 metrics, carrying no body-measurement or photo content into events. Same environment tagging as Sentry.

NFR-015 — An E2E test automation baseline is established. Priority flows: staff provisioning + role enforcement, the payment cutover path, progress-data access boundaries, and class booking capacity limits. Complements, not replaces, V1.0 CI gates.

NFR-016 — Coach access to member progress data is re-verified on every request against current assignment state; an ended assignment revokes read access with no caching window that outlives the assignment.

NFR-017 — Per-gym Tara Money credentials (FR-126) are stored encrypted at rest, accessible only to the server-side payment service, never returned to any client, never logged, never readable across tenants. They carry the same isolation guarantees as all tenant data (NFR-001).

NFR-018 — Tenant suspension for SaaS non-payment (FR-131/FR-132) is enforced at the authorization layer, not only the UI: a suspended gym's staff and members are denied at the RLS/auth-hook layer, so suspension cannot be bypassed by a client ignoring UI state. Takes effect on the next request; no tenant data is deleted or mutated.

8. Out of Scope — V1.5

Deferred beyond V1.5 (nothing added without a recorded decision in .decision-log.md):

Item	Target Version
Travel mode plan type	V2.0
Streaks, challenges, leaderboards, activity feed	V2.0
Coach-to-receptionist escalation for expiring clients	V2.0
Class waitlists (auto-promote when a spot frees)	V2.0
Provider-executed refund API calls	V2.0
Nutrition / meal logging	V2.0
Progress data export (member downloads history)	V2.0
Live multi-currency payments beyond XAF	V2+
Gym-owned merch store	V2.5
Platform-wide marketplace	V3.0
Geofence / Bluetooth / WiFi presence detection	V3.0
Manager/Owner visibility into member progress data	Not planned (privacy model)
Per-gym App Store listings; custom fonts / themes	V4+
AI features, wearables, corporate wellness	V4+

Explicit non-goal: V1.5 does NOT open member body/progress data to gym management. The member + assigned-coach boundary is a product commitment, not a temporary limitation. Any future change requires a recorded decision and member consent handling.

9. Open Questions

#	Question	Owner	Blocker for
OQ-7	Tara Money sandbox spike — exit criteria per FR-100; outcome in docs/decisions.md	Dev team	Payments cutover
OQ-8	Progress photo storage cost/retention at beta scale; confirm Storage limits + retention policy	Dev team	FR-095 / NFR-011
OQ-9	Which V1.0 plan types beyond Class-only grant class booking? Owner-configurable or fixed?	Product	FR-105
OQ-10	Confirm default measurement fields for the Cameroon beta cohort	Product	FR-094
OQ-11	Do beta gyms need co-owners (>1 Owner per gym)? V1.5 assumes one Owner	Product	FR-087 scope
OQ-12	WhatsApp/Evolution API compliance & number provisioning; messaging limits & sender identity	Dev team	FR-117–FR-119
OQ-13	Tara Money API: confirm create-collect + payment-detection flow (callback vs poll) and whether payments settle into each gym's own Tara account (per-gym credentials/sub-accounts).	Dev team	FR-124, 126, 128
OQ-14	Tara Money recurring/subscription support for Flow B — automated recurring debits, or a fresh collect each SaaS cycle the owner approves?	Dev team	FR-130, 133
OQ-15	SaaS billing proration on mid-cycle tier change — prorate or apply next cycle?	Product	FR-130

10. Release Definition — “Beta-Ready”

V1.5 is beta-ready when a gym the founding team has not personally hand-held can:

1. Be created by GymOS (founder-onboarded), then staff itself — add its own manager, receptionist, and coach — with no support ticket (FR-087–FR-092).
2. Take real payments through Tara Money, with the cutover proven clean (FR-099–FR-103, NFR-012).
3. Run the retention loop end to end: front-desk alert (V1.0) plus members returning to the app between visits to log progress (FR-093–FR-098).
4. Operate its weekly rhythm — classes scheduled and booked, workout plans assigned (FR-104–FR-112).
5. Do all of the above bilingually, offline-tolerant where FR-097 requires, with PostHog showing whether the loops work and E2E tests guarding critical paths (NFR-014, NFR-015).

Everything in Section 8 is deliberately out, so the beta ships on time rather than growing into V2.